

ENGI 9875 Project Proposal

Predictive Maintenance with Anomaly Detection

By Onwukanjo Innocent (202487225)

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1. Project Scope

This project proposes a predictive maintenance system designed to detect equipment failure before it happens by monitoring simulated motor vibration data for anomalies, detects anomalous behavior, and triggers an alert before a failure occurs. The entire system will be developed in C running on the NuttX RTOS, targeting the STM32F407 platform and simulated using Renode.

Because physical boards are limited, pre-recorded vibration data will be used. The anomaly detection relies on an autoencoder neural network trained offline in Python. This trained model will then be converted into a C-compatible format and integrated into NuttX as an inference task. The system is structured as a pipeline of separate NuttX tasks responsible for periodic sensor sampling, realtime inference, anomaly scoring, event logging, and alert generation.

To manage the realtime embedded environment, the project would utilize specific NuttX services. Rate monotonic scheduling would be applied to the periodic sampling task, which would have the highest priority. Message buffers would transfer data between the sampling and inference tasks and counting semaphores would manage concurrent write access to the shared anomaly event log. Task notifications would directly wake the low priority alert tasks when an anomaly is detected.

The primary focus of this project is to gain practical experience with task scheduling, inter-task communication, and concurrency management in NuttX, while also exploring how a machine learning model can be deployed within a real-time embedded environment.

2. RTOS Services Used

The following NuttX services are planned for this project:

- **Periodic Sampling Task:** A task that reads vibration data at a fixed interval. Rate-monotonic scheduling will be applied, and deadline feasibility will be analyzed.
- **Message Buffers:** Used to transfer vibration data from the sampling task to the inference task, allowing both to run independently without blocking each other.
- **Counting Semaphores:** Applied to manage concurrent write access to the shared anomaly event log across multiple tasks.
- **Task Notifications:** Used to wake the alert task directly when the inference task identifies an anomaly, avoiding unnecessary polling.

- **Priority Assignment:** Tasks will be assigned priorities based on their timing requirements. The sampling task will hold the highest priority, and the logging task the lowest.

3. Hardware Requirements

The target platform is the STM32F407. Development and demonstration will be carried out entirely in Renode.

Some other tools we will use are:

- Python + Keras: for offline autoencoder training and vibration dataset generation
- TensorFlow Lite Micro: for running the trained model as an inference task inside NuttX

4. Learning Outcomes Addressed

- **LO1: Real-Time Scheduling:** Rate-monotonic scheduling will be applied to the sampling task, with schedulability analysis performed to confirm deadlines are met.
- **LO3: Concurrency Issues:** Shared access to the anomaly log across tasks will provide a practical scenario for identifying and resolving concurrency issues using semaphores.
- **LO5: Device Drivers:** A simulated accelerometer driver will be written as a NuttX character device in C, covering the driver interface and its integration with the task layer.